

BloodLink — Product Requirements Document (PRD)

AI-Powered Emergency Blood Coordination Platform

1. Product Overview

Product Name: BloodLink

Product Type: AI-powered emergency blood coordination platform

Domain: Healthcare

BloodLink helps hospitals and authorized users quickly find suitable blood donors during emergencies. It intelligently matches blood requests with nearby eligible donors using blood compatibility, distance, availability, urgency, and response behavior. The platform also provides blood-demand analytics and can identify possible future blood shortages.

2. Problem Statement

During accidents, surgeries, and other medical emergencies, finding compatible blood quickly can be difficult. Existing donor platforms may mainly depend on blood-group searches and may not effectively prioritize donors according to:

- Blood compatibility
- Distance from the hospital
- Donor availability
- Emergency urgency
- Likelihood of responding

Blood banks may also experience shortages because future demand is difficult to predict.

3. Product Goals

- Reduce the time required to find suitable blood donors.
- Connect verified emergency blood requests with suitable donors.
- Prioritize suitable donors instead of notifying everyone.
- Provide location-based donor matching.
- Track blood demand and availability.
- Identify possible blood shortages.
- Reduce fake or unauthorized emergency requests.
- Improve coordination between hospitals, donors, and blood banks.

4. Target Users

Primary Users

- Patients requiring blood
- Registered blood donors
- Hospitals and emergency departments

- Blood banks and transfusion centers

Secondary Users

- Emergency-response coordinators
- Healthcare administrators
- Blood-bank administrators

5. Core Features

5.1 User Authentication

Register/login with role-based access for Donor, Hospital, Blood Bank, and Administrator.

5.2 Donor Profile

Maintain blood group, approximate location, availability, last donation date, donation history, and response history. Sensitive information must remain protected.

5.3 Emergency Blood Request

Authorized hospitals create requests with blood group, units, hospital, location, emergency level, required time, notes, and status.

5.4 AI Donor Matching

Rank eligible donors using blood compatibility, distance, availability, urgency, and response behavior. Produce an explainable 0–100 AI-assisted match score.

5.5 Blood Compatibility

Centralized reusable compatibility service supporting A+, A–, B+, B–, AB+, AB–, O+, and O–. Final eligibility and transfusion decisions remain with qualified professionals.

5.6 Location-Based Matching

Use OpenStreetMap and Leaflet to display hospital/request locations, nearby donors, distances, and blood banks while protecting exact donor locations.

5.7 Smart Notifications

Use an in-app notification system for the hackathon, with Firebase Cloud Messaging prepared as an optional extension. Track sent, viewed, accepted, declined, and expired events.

5.8 Donor Chain

If a donor declines, does not respond, or becomes unavailable, automatically move to the next suitable donor and show the chain in the hospital dashboard.

5.9 Blood Inventory

Blood banks manage available units for all supported blood groups and update inventory.

5.10 Blood Shortage Prediction

Analyze historical demand by blood group, date, region, units, and emergency level to estimate expected demand and shortage risk.

5.11 Analytics Dashboard

Display donors, requests, inventory, demand trends, regional demand, matching time, donor distance, fulfillment, and shortage risk.

5.12 Admin Management

Verify hospitals/blood banks, manage users and requests, review suspicious activity, and view audit logs.

6. System Workflow

Hospital → Create Request → Validate → Verify → Find Compatible Donors → Check Availability → Calculate Distance → AI Ranking → Notify Donors → Donor Response → Accept or Continue Donor Chain → Fulfill Request → Update Analytics

7. AI/ML Requirements

A. Donor Matching

- Filter incompatible donors.
- Filter unavailable donors.
- Calculate approximate distance.
- Consider emergency urgency.
- Consider donor response history.
- Generate configurable 0–100 matching score.
- Provide an explanation for the recommendation.

B. Blood Shortage Prediction

- Use historical demand data.
- Consider blood group and region.
- Analyze time trends and inventory.
- Return expected demand, shortage risk, and recommendation.
- Use synthetic/demo data when real data is unavailable.

The matching engine should be explainable and should not claim to be a medically validated model.

8. Technology Stack

Layer	Technology
AI Development	Z.ai
Frontend	Next.js, React, TypeScript
Styling	Tailwind CSS
Backend	Next.js API Routes / Server Actions
Database	Neon PostgreSQL
ORM	Prisma
Deployment	Vercel
Authentication	Auth.js or secure Next.js-compatible authentication
Maps	OpenStreetMap + Leaflet
AI/ML	Explainable matching engine; modular ML/prediction service
Notifications	In-app notifications; optional Firebase Cloud Messaging

Layer	Technology
Analytics	Snowflake + Streamlit (optional/advanced)
Version Control	GitHub

Architecture principle: Neon PostgreSQL is the primary transactional database. Snowflake, if configured, is used mainly for analytics and historical workloads and is not required for the core application.

9. Functional Requirements

- FR1** — User registration and authentication.
- FR2** — Role-based access for donor, hospital, blood bank, and administrator.
- FR3** — Donor profile management.
- FR4** — Hospital and blood-bank profile management.
- FR5** — Emergency blood request creation and tracking.
- FR6** — Blood compatibility filtering.
- FR7** — AI-assisted donor ranking.
- FR8** — Location-based donor matching.
- FR9** — Donor accept/decline functionality.
- FR10** — Automatic donor-chain fallback.
- FR11** — Emergency in-app notifications.
- FR12** — Blood inventory management.
- FR13** — Blood-demand analytics.
- FR14** — Shortage-risk prediction.
- FR15** — Admin verification and management.

10. Non-Functional Requirements

Performance

Donor matching should complete quickly during emergency requests.

Security

Protect sensitive donor and healthcare information with secure authentication, authorization, validation, and server-side access controls.

Scalability

Support increasing numbers of donors, hospitals, requests, and analytics records.

Reliability

Emergency requests must be reliably stored and tracked.

Usability

The interface must remain simple and clear during stressful emergency situations.

Privacy

Do not expose exact donor locations or unnecessary personal information.

11. Hackathon Demo Scenario

Hospital creates an O– emergency request for 2 units with CRITICAL urgency and a 2-hour requirement.

- System finds compatible donors.
- Donor A: O–, 1.8 km, available, Match Score 94.
- Donor B: O–, 3.4 km, available, Match Score 87.
- Donor C: O+, 2.1 km, not compatible.
- Donor A declines.
- System automatically moves to Donor B.
- Donor B accepts.
- Request becomes DONOR FOUND.
- Dashboard then shows blood-demand analytics and an O– shortage prediction.

12. Success Metrics

- Time required to find a donor
- Number of compatible donors identified
- Average donor distance
- AI match score
- Donor response rate
- Request completion rate
- Donor-chain success
- Shortage prediction accuracy using demo data
- Reduction in unnecessary notifications

13. Healthcare Safety & Privacy

BloodLink is an AI-assisted coordination and decision-support platform. It does not diagnose patients, determine final donor eligibility, guarantee blood compatibility, or replace qualified healthcare professionals or authorized blood banks. All transfusion decisions and compatibility testing must be performed by qualified professionals.

Do not use real patient medical information in the hackathon prototype. Use synthetic/demo data.

14. Expected Outcome

- Reduce donor-search time.
- Improve emergency coordination.
- Prioritize relevant donors.
- Reduce unnecessary notifications.
- Improve blood-bank preparedness.
- Identify possible shortages earlier.
- Improve access to blood during emergencies.

15. One-Line Product Pitch

“BloodLink uses AI and real-time location to connect emergency blood requests with the most suitable donors while predicting future blood shortages.”